



Operations Research

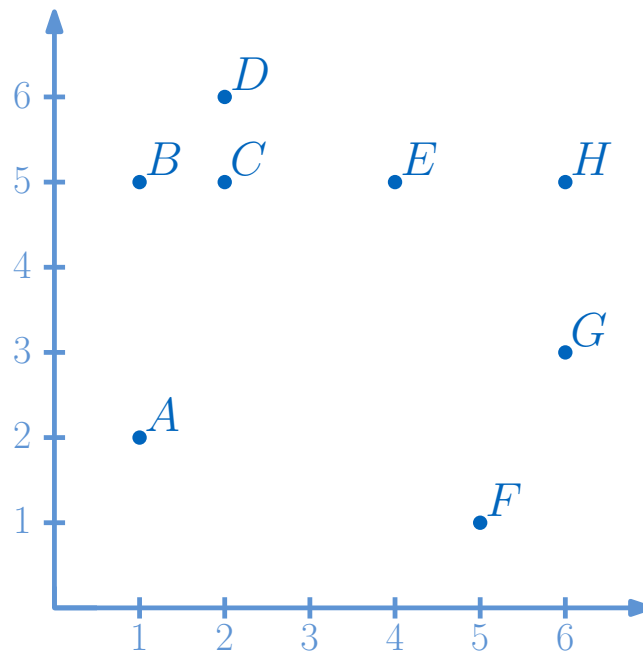
Prof. Dr. Martin Bichler,
Halil Bayrak, Janik Bürgermeister, Teodora Dobos,
Jan Höhener, Abheek Ghosh

Exercise Sheet 9

Demo Exercises

Exercise D9.1. *TSP-Approximation*

Consider the points A to H in the plane:



The distance between $p_1 = (x_1, y_1) \in \mathbb{R}^2$ and $p_2 = (x_2, y_2) \in \mathbb{R}^2$ is given by the euclidean norm: $d(p_1, p_2) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$.

- Determine a minimum spanning tree using Kruskal's algorithm. List the edges in the order they are added by the algorithm.
- Provide the tour you obtain when you execute the 2-approximation algorithm of Christofides using your spanning tree from part (a). Start your tour at node A .
- What is the length of the tour you obtained and which lower bound can you give for the optimal tour? Where does the approximation factor of 2 come from?

Exercise D9.2. *Collect electric scooters*

Fabienne and Mattea work for the electric scooter operator *boi* and are tasked with collecting empty scooters and taking them to a depot. Fabienne is stationed in depot 1 (D_1), while Mattea is in depot 2 (D_2). The two are given a list of locations $S_1, \dots, S_n$ with empty scooters that they should collect ($n \geq 4$). The cost of a trip from node i to node j for $i, j \in V$ is denoted by c_{ij} . Let $V = \{S_1, \dots, S_n, D_1, D_2\}$ be the set of all nodes and let the variables x_{ij}^F or x_{ij}^M be whether Fabienne or Mattea uses the line segment between $i, j \in V$ for $i \neq j$. Fabienne and Mattea want to model the problem at hand as an integer linear program and determine the minimum effort for both.

Formulate the following constraints. If you introduce new variables, also provide a brief description.

- a) Each scooter location must be visited exactly once.
- b) Fabienne and Mattea leave their respective depots exactly once and also return exactly once.
- c) Mattea and Fabienne each do a round trip.
- d) Finally, formulate an appropriate objective function.

Exercise D9.3. *NP-hardness of TSP*

The decision problem of whether a Hamiltonian cycle (HC) exists in an undirected and unweighted graph $G = (V, E)$ is generally NP-complete. Show that the *metric Traveling Salesperson Problem* (TSP) is at least as hard as the HC problem.

To do this, show that you can represent any HC problem as a solution to a specific TSP. In doing so, you will have demonstrated a *reduction* from the HC problem to the TSP problem.